

Program 10:

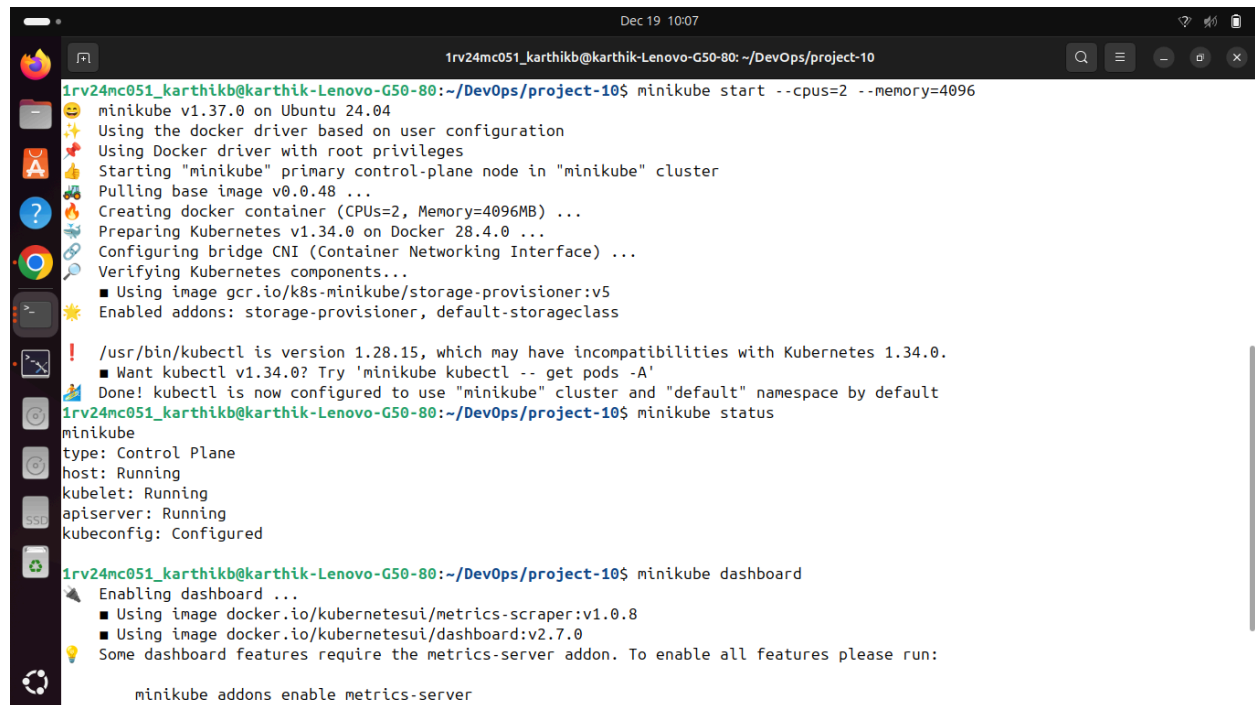
Project Structure: project-10

->deployment.yaml

->service.yaml

Step 1: Create deployment.yaml

Step 2: Create service.yaml

A terminal window titled '1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/project-10' showing the execution of minikube commands. The 'minikube start' command is run with flags '--cpus=2' and '--memory=4096'. The output shows the process of starting a minikube cluster on Ubuntu 24.04 using Docker. It includes steps like pulling the base image, creating a Docker container, preparing Kubernetes v1.34.0, and configuring the bridge CNI. A warning message indicates that the installed kubectl version (1.28.15) is incompatible with the Kubernetes version (1.34.0) and suggests using 'minikube kubectl -- get pods -A'. The 'minikube status' command is then run, showing that the minikube control plane, host, kubelet, apiserver, and kubeconfig are all running or configured. Finally, the 'minikube dashboard' command is run, which enables the dashboard and suggests using specific images for the metrics-scraper and dashboard. A note at the bottom states that some dashboard features require the metrics-server addon and provides the command 'minikube addons enable metrics-server'.

Step 3: Apply deployment & service

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80: ~/DevOps/project-10
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~$ cd DevOps/
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ cd project-10
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ ls
deployment.yaml  service.yaml
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl apply -f service.yaml
service/nginx-service created
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-28jkk   0/1     ContainerCreating   0           33s
nginx-deployment-6f9664446b-qvcmk   0/1     ContainerCreating   0           33s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-28jkk   1/1     Running    0           58s
nginx-deployment-6f9664446b-qvcmk   1/1     Running    0           58s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl get servies
error: the server doesn't have a resource type "servies"
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl get svc
NAME      TYPE      CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes  ClusterIP  10.96.0.1     <none>        443/TCP   5m20s
nginx-service  NodePort   10.101.108.119 <none>        80:32155/TCP 93s
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  5m36s  v1.34.0
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl logs nginx-deployment-6f9664446b-28jkk
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
```

Step 4: Verify resources

Step 5 :Describe and view details

```
2025/12/19 04:15:59 [notice] #1: start worker process 32
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ kubectl describe pod nginx-deployment-6f9664446b-28jkk
Name:                               nginx-deployment-6f9664446b-28jkk
Namespace:                           default
Priority:                             0
Service Account:                       default
Node:                                 minikube/192.168.49.2
Start Time:                           Fri, 19 Dec 2025 09:45:02 +0530
Labels:                               app=nginx
                                      pod-template-hash=6f9664446b
Annotations:                           <none>
Status:                               Running
IP:                                   10.244.0.6
IPs:                                  IP: 10.244.0.6
Controlled By:                         ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:   docker://756944442a7ad3a465dc277f48f6db74369e8b8622753aca4a0f322469f12e0c
    Image:          nginx:latest
    Image ID:       docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Fri, 19 Dec 2025 09:45:59 +0530
    Ready:          True
    Restart Count:   0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-8gchd (ro)
Conditions:
```

Step 5 :Access the service

```

Normal        11m    kubelet    Successfully pulled image nginx:latest in 3.577s (30.473s including waiting). If
ge size: 151914114 bytes.
Normal        Created    11m    kubelet    Created container: nginx
Normal        Started    11m    kubelet    Started container nginx
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ minikube service nginx-service

```

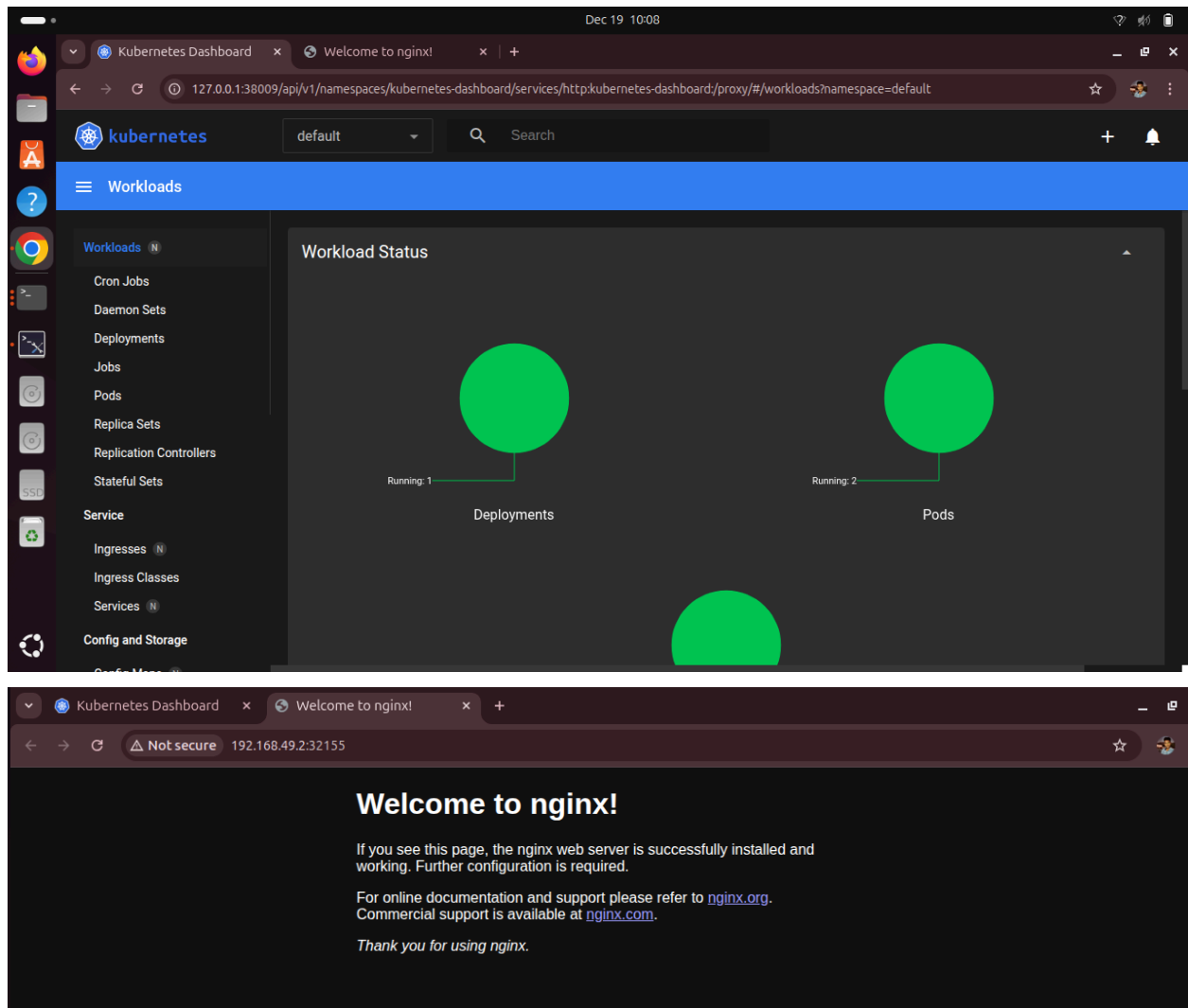
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32155

```

🔥 Opening service default/nginx-service in default browser...
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$ Opening in existing browser session.
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/project-10$

```

Step 6 :Open Kubernetes Dashboard



The image shows two browser windows. The top window displays the Kubernetes Dashboard at the URL `127.0.0.1:38009/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard/proxy/#/workloads?namespace=default`. The dashboard shows a 'Workload Status' section with two green circles representing 'Deployments' (Running: 1) and 'Pods' (Running: 2). The left sidebar lists various Kubernetes resources like Cron Jobs, Daemon Sets, Deployments, Jobs, Pods, Replica Sets, Replication Controllers, Stateful Sets, Services, Ingresses, and Config and Storage.

The bottom window shows the Nginx welcome page at the URL `192.168.49.2:32155`. The page has a dark background with the text 'Welcome to nginx!' in white. Below the title, it states: 'If you see this page, the nginx web server is successfully installed and working. Further configuration is required.' It also provides links for online documentation (nginx.org) and commercial support (nginx.com), and ends with 'Thank you for using nginx.'